DDTCDR: Deep Dual Transfer Cross Domain Recommendation

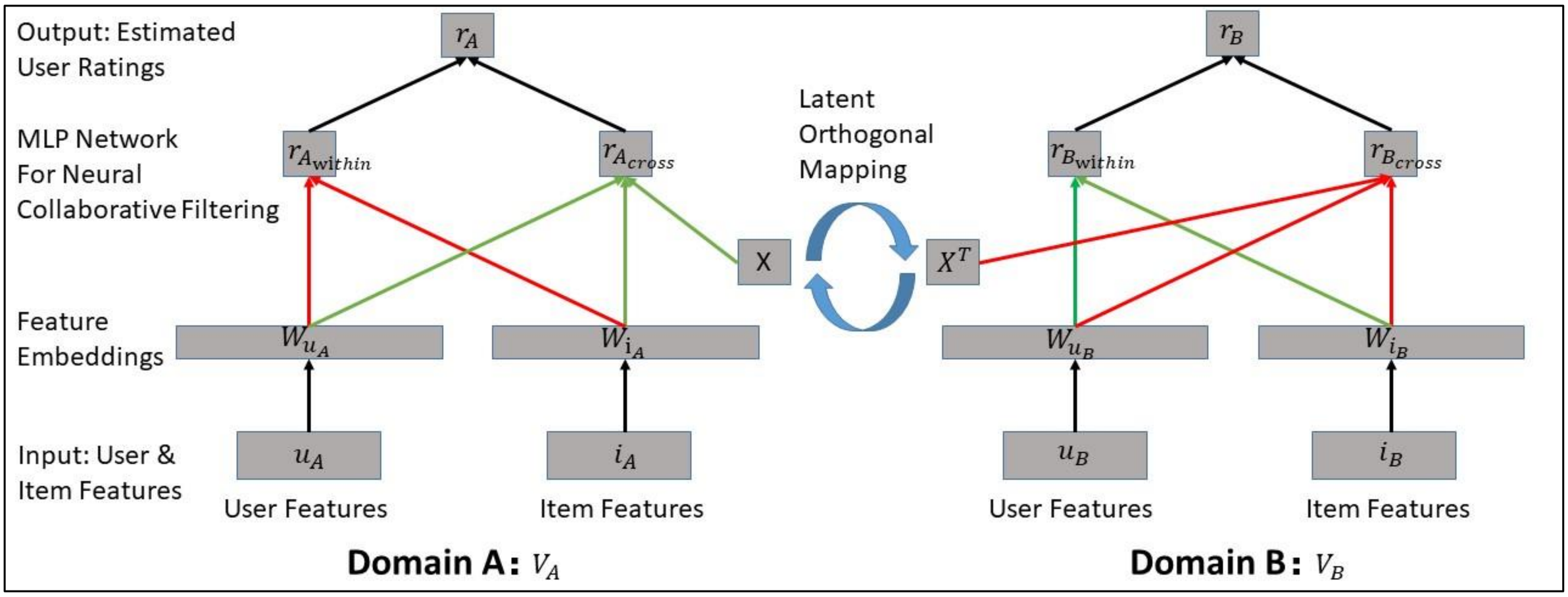
Pan Li, Alexander Tuzhilin
New York University, Stern School of Business
pli2@stern.nyu.edu, atuzhili@stern.nyu.edu

Problem

Previous cross-domain recommendation models do not

- utilize bidirectional transfer learning mechanism (and use only unidirectional transfer learning) to provide cross-domain recommendations
- specify orthogonality of the mapping function between different domains
- improve the recommendation models of both domains simultaneously through training process

Solution: DDTCDR



DDTCDR consists of three components:

1. **Feature AutoEncoding** that maps user and item features into latent embeddings through MLP network

$$W_{u_A} = AE_A(u_A),\ W_{u_B} = AE_B(u_B),\ W_{i_A} = AE_A(i_A),\ W_{i_B} = AE_B(i_B)$$

$$L = ||u_a - MLP_{dec}(MLP_{enc}(u_a))||$$

2. **Latent Orthogonal Mapping *X*** that learns an orthogonal mapping between user embeddings W_{u_a}, W_{u_b} across different domains
   - Preserves similarities between user embeddings across different latent spaces
   - Automatically derives the inverse mapping function
3. **Deep Dual Transfer Learning** that transfers user preferences from one domain to the other bidirectionally and provide rating estimation accordingly.

Repeat

$$r'_A = (1-\alpha)RS_A(W_{u_A}, W_{i_A}) + \alpha RS_B(X * W_{u_A}, W_{i_A})$$

$$r'_B = (1-\alpha)RS_B(W_{u_B}, W_{i_B}) + \alpha RS_A(X^T * W_{u_B}, W_{i_B})$$

Until Convergence

Experiments

Data

- Online recommendation service for books, movies and music

Features

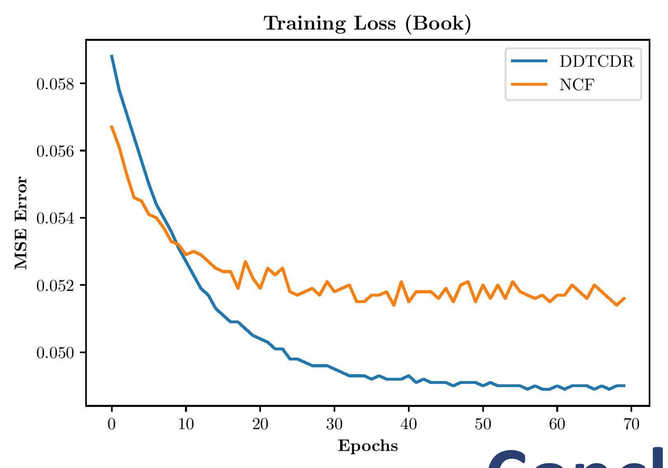
- User (Gender, Age, Movie Taste, Residence, Preferred Category, Recommendation Usage, Marital Status, Personality)
- Book (Category, Title, Author, Publisher, Language, Country, Price, Date)
- Movie (Genre, Title, Director, Writer, Runtime, Country, Rating, Votes)
- Music (Listener, Playcount, Artist, Album, Tag, Release, Duration, Title)

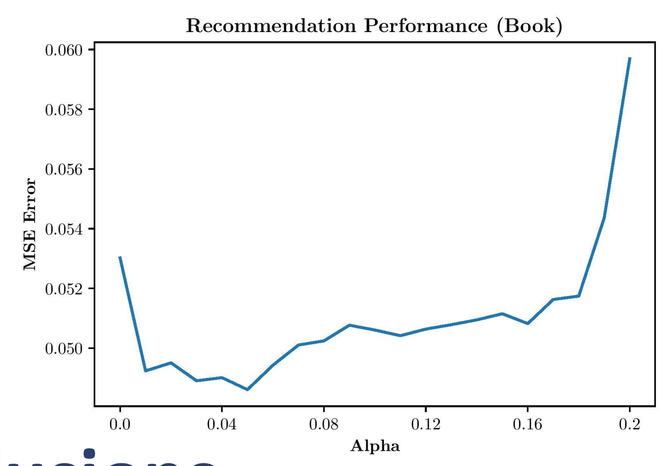
Baselines (compared with our DDTCDR model)

- NCF • CMF • CoNet • CDFM • CCFNet

Results

Algorithm	Book				Movie			
	RMSE	MAE	Precision@5	Recall@5	RMSE	MAE	Precision@5	Recall@5
DDTCDR	**0.2213***	**0.1708***	**0.8595***	**0.9594***	**0.2213***	**0.1714***	**0.8925***	**0.9871***
Improved %	(+3.98%)	(+9.54%)	(+2.77%)	(+6.30%)	(+2.44%)	(+9.80%)	(+2.75%)	(+2.74%)
NCF	0.2315	0.1887	0.8357	0.8924	0.2276	0.1895	0.8644	0.9589
CCFNet	0.2639	0.1841	0.8102	0.8872	0.2476	0.1939	0.8545	0.9300
CDFM	0.2494	0.2165	0.7978	0.8610	0.2289	0.1901	0.8498	0.9312
CMF	0.2921	0.2478	0.7972	0.8523	0.2738	0.2293	0.8324	0.9012
CoNet	0.2305	0.1892	0.8328	0.8990	0.2298	0.1903	0.8680	0.9601





Conclusions

- **Dual Transfer Learning mechanism significantly helps to improve cross-domain recommendation performance.**
- **Orthogonality method plays an important role in learning the mapping function across different domains**